



Dynamic Pricing Strategy for Revenue Optimization

DATA-DRIVEN APPROACH USING DEMAND FORECASTING &
PRICE ELASTICITY

PRESENTED BY: SUMEDHA PATIL



Problem Statement

- Fixed pricing structure for all classes
 - No demand-based price adjustments
 - Peak sessions overcrowded
 - Off-peak sessions underutilized
 - Revenue not optimized



Project Objectives

- Identify high-demand classes and time slots
 - Analyze price elasticity
 - Forecast future demand
 - Develop dynamic pricing algorithm
 - Increase revenue per class and overall revenue



Data Analysis Performed

- Exploratory Data Analysis (EDA)
 - Price Elasticity Modeling
 - Time Series Demand Forecasting
 - Correlation Analysis
 - Revenue Simulation



Key Insights

- Evening sessions show highest demand
 - Weekends have higher bookings
 - HIIT and Yoga are most popular classes
 - Afternoon sessions show lower utilization
 - Peak demand less price-sensitive

Price Elasticity Findings

- Negative correlation between price and demand
 - Elastic demand in some classes
 - Inelastic demand during peak hours
 - Opportunity to increase price during peak



Demand Forecasting Results

- Consistent peak demand in evenings
 - Stable high weekend demand
 - Recurring patterns across months
 - Supports premium pricing during peak periods



Dynamic Pricing Strategy

- Increase price during peak hours and weekends
 - Discount prices during off-peak sessions
 - Adjust price based on demand intensity
 - Incorporate elasticity into pricing formula

Dynamic Pricing Formula

- Dynamic Price = Base Price ×
 - (1 + Demand Factor - Elasticity Adjustment)
 - Demand Factor based on booking rate
 - Elasticity adjustment controls price sensitivity



Revenue Simulation

- Calculated revenue before dynamic pricing
 - Simulated revenue after dynamic pricing
 - Observed measurable revenue improvement
 - Improved utilization during low-demand periods



Business Benefits

- Higher revenue per peak class
 - Better capacity utilization
 - Balanced demand distribution
 - Data-driven strategic decisions



Conclusion & Future Scope

- Dynamic pricing improves revenue efficiency
 - Supports long-term business growth
 - Future: ML-based real-time pricing
 - Scalable to multiple locations